

Question	Answer	Mark	Mark scheme	Additional guidance
19	2030	P1 P1 P1 A1	for process to find diameter of base of small cone, eg 16 seen as diameter or 8 seen as radius or for scale factor (length) is $\frac{2}{3}$ oe or $\frac{3}{2}$ oe for process to find the volume of one cone, eg $\frac{1}{3} \times \pi \times 12^2 \times 15$ ($= 720\pi$ or 2261.946...) or $\frac{1}{3} \times \pi \times "8"^2 \times 10$ ($= \frac{640}{3}\pi$ or 670.206...) or for process to find the volume of the hemisphere, eg $\frac{2}{3} \times \pi \times 12^3$ ($= 1152\pi$ or 3619.114...) or for scale factor (volume) is $\left(\frac{2}{3}\right)^3$ oe for full process to find volume of frustum and hemisphere, eg "2261.946..." – "670.206..." ($= \frac{1520}{3}\pi$ or 1591.740...) or "2261.946..." $\times \left(1 - \left(\frac{2}{3}\right)^3\right)$ and $\left(\frac{4}{3} \times \pi \times 12^3\right) \div 2$ oe ($= 1152\pi$ or 3619.114...) for answer in the range 2026 to 2030	If an answer is shown in the range in working and then incorrectly rounded award full marks.